

ARUNABHA NAG

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PROFESSIONAL SUMMARY

Results-driven Computer Science Engineering undergraduate with hands-on experience designing and deploying production-grade MERN applications with secure JWT authentication, RESTful APIs, Chrome extensions, and LLM-heuristic pipelines. Proficient in integrating Google Gemini models, optimizing database schemas, and applying OWASP-aligned defensive security protocols. National-level technical hackathons including Smart India Hackathon Top 10 and IIT Bombay National Entrepreneurship Challenge AIR 5 (Finalists).

PROJECTS

Verdict Engine | Enterprise Review Analysis & Scoring Pipeline

Full-Stack Developer (TypeScript, React, Chrome Extension, Gemini API) | [GitHub](#)

- Balanced Scraping Engine:** Built a multi-threaded background scraper that fetches 10 pages in parallel (4 recent, 3 critical, 3 oldest) to pull 100 consolidated reviews for quality analysis.
- Heuristic-LLM Hybrid Filter:** Implemented a priority scoring system that ranks the 100 scraped reviews, sending only the top 25 high-priority defect reviews + 5 general reviews to Gemini (reducing API token costs by 70%), and backfilling the remaining 70 using rating-heuristic mappings.
- Quantitative Scoring & Math Engine:** Programmed a scoring service calculating authenticity indicators and Tier classifications, reducing full 100+ review analysis latency to under 3 seconds.

WealthifyMe | Personal & Family Finance Tracker

Full-Stack Developer (MERN Stack, Recharts, Stripe) | [Live Link](#) |

- AI Financial Assistant:** Integrated the gemini-3.5-flash model to analyze historical transaction records and generate tailored savings plans, automating expense categorization with 85%+ precision.
- Dual-Layer Categorization:** Implemented a live Gemini classification pipeline with a fallback regex engine containing 11 distinct financial rules (mapping queries to categories like Food, Rent, Transport, and Investments).
- Shared Workspace & Warning Systems:** Built a collaborative household budgeting module supporting multi-user workspace tracking, featuring automated warnings when joint expenses exceed 80% and 100% thresholds.
- Stripe Webhook Integration:** Implemented Stripe billing webhooks to securely coordinate tier-progression logic (Free to Pro plan) with zero database state inconsistencies.

Secure MERN Portfolio & Admin Portal

Full-Stack Developer (MERN, JWT Auth) | [Live Link](#) |

- Defensive Security Protocols:** Designed and deployed a secure portfolio site implementing Helmet security headers, IP rate-limiting, and XSS sanitization on form inputs to neutralize NoSQL injections and scripting vulnerabilities.
 - OWASP Alignment:** Configured Node/Express backend parser to enforce a strict 10KB body payload limit to prevent memory exhaustion, and implemented rate-limiting capped at 100 requests per 15 minutes per IP.
 - Dynamic Admin Dashboard:** Constructed a CRUD Admin Portal (/admin) protecting timeline metrics, skills tracks, and inquiry logs with short-lived 2-hour JWT sessions and bcrypt security question recovery.
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KEY SKILLS

- HTML5
 - CSS3
 - Data Structures and Algorithm
 - Javascript
 - React
 - Node.js
 - MongoDB
 - MySQL
 - TypeScript
 - VERCEL
 - RENDER
 - Express.js
 - python
 - Microsoft Excel
 - Microsoft Access
 - Microsoft Powerpoint
 - Aptitude
 - JWT
 - Git/Github
 - Redis
 - Microsoft Azure
 - Google Cloud Platform
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EDUCATION

School Of Engineering And Technology, H.N.B.G.U. Central University (Sep 2023 - Current)

Bachelor of Technology - Computer Science and Engineering, SGPA: 6.5/10.0

Srinagar , Garhwal, Uttarakhand

Aditya Academy Senior Secondary , Central Board Of Secondary Education (CBSE)

Higher Secondary - Science Group - (Maths , Physics , Chemistry , Biology), 82.8% (Aggregate)

Kolkata, West Bengal

ACHIEVEMENTS

- 1. Technical & Research Lead, Team Stratovators (IIT Bombay NEC 2025): Secured All India Rank (AIR) 5 out of 200+ competing engineering teams at the National Entrepreneurship Challenge.**
 - 2. Team Lead (Smart India Hackathon 2024): Selected in the Top 10 teams globally for automated software solutions.**
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LANGUAGES

- English (Proficient)
- Hindi (Proficient)
- Bengali (Native/ Mother Tongue)